16.6.2026 Pauli Partinen

# Fingrid’s Perspective on Data Center Modelling Requirements and Model Quality

ESIG 2026 Summer Workshops, Large Load Modeling, Testing and Interconnection Requirements Workshop, June 16, Denver

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# Content

1. Introduction to Fingrid
2. How to get models?
3. What to model?
4. How to make sure models work?

# Fingrid is Finland's transmission system operator.

**We secure reliable electricity for our customers, and we shape the power system of the future.**

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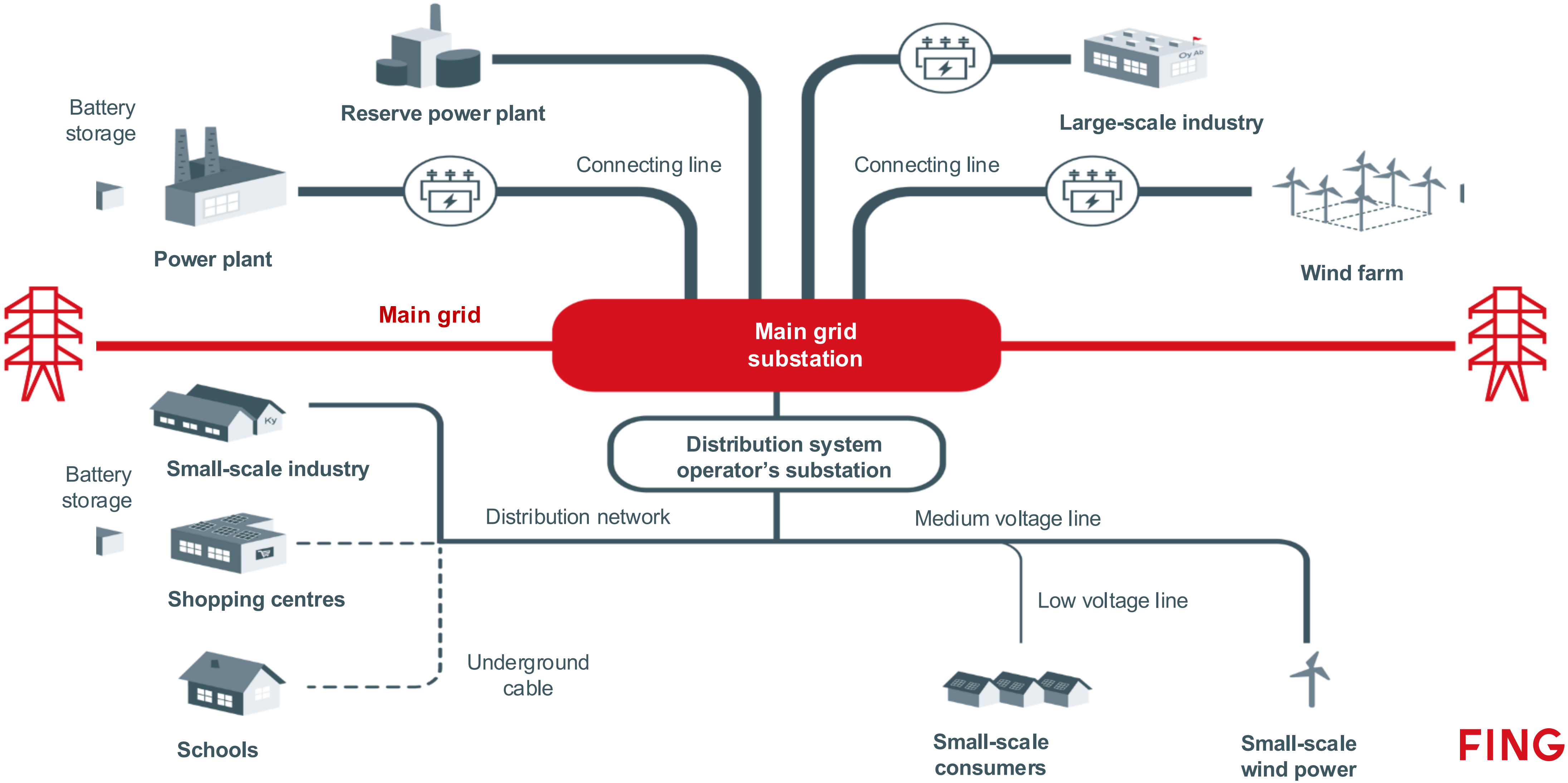
# Electricity system of Finland

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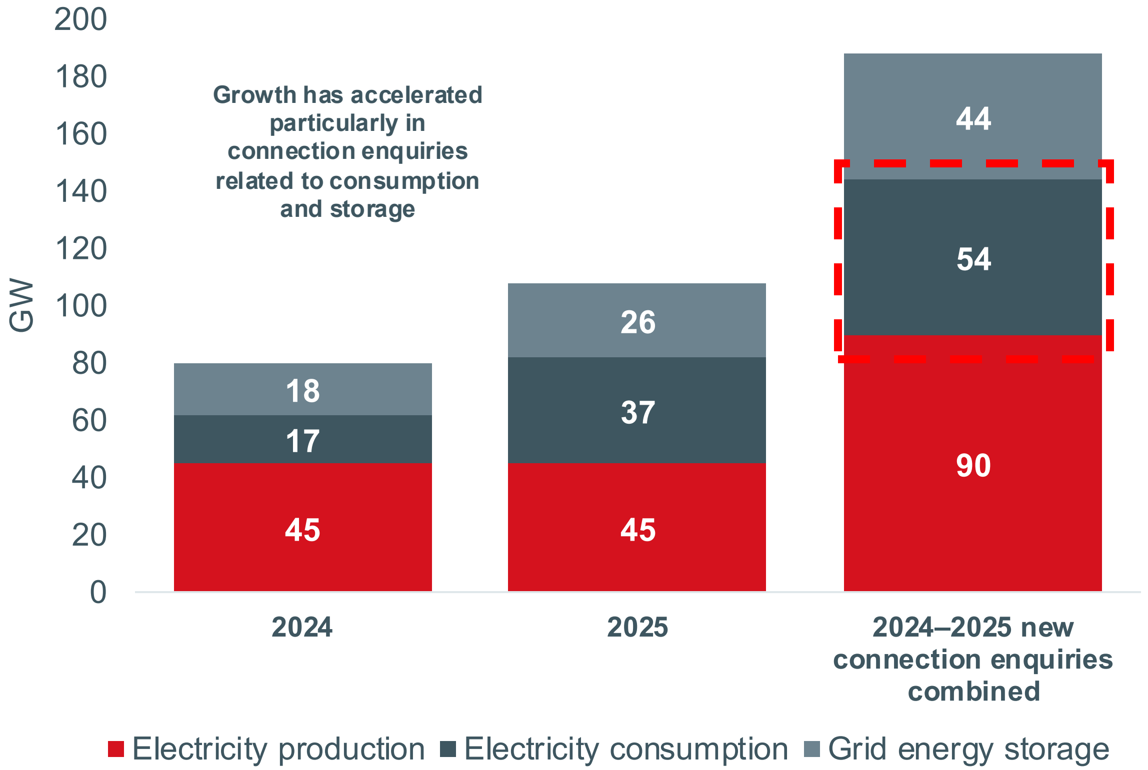
# Clean electricity attracts electricity-intensive industry

**New main grid connection enquiries* during the year**




*Connection enquiries are non-binding and based on preliminary project information

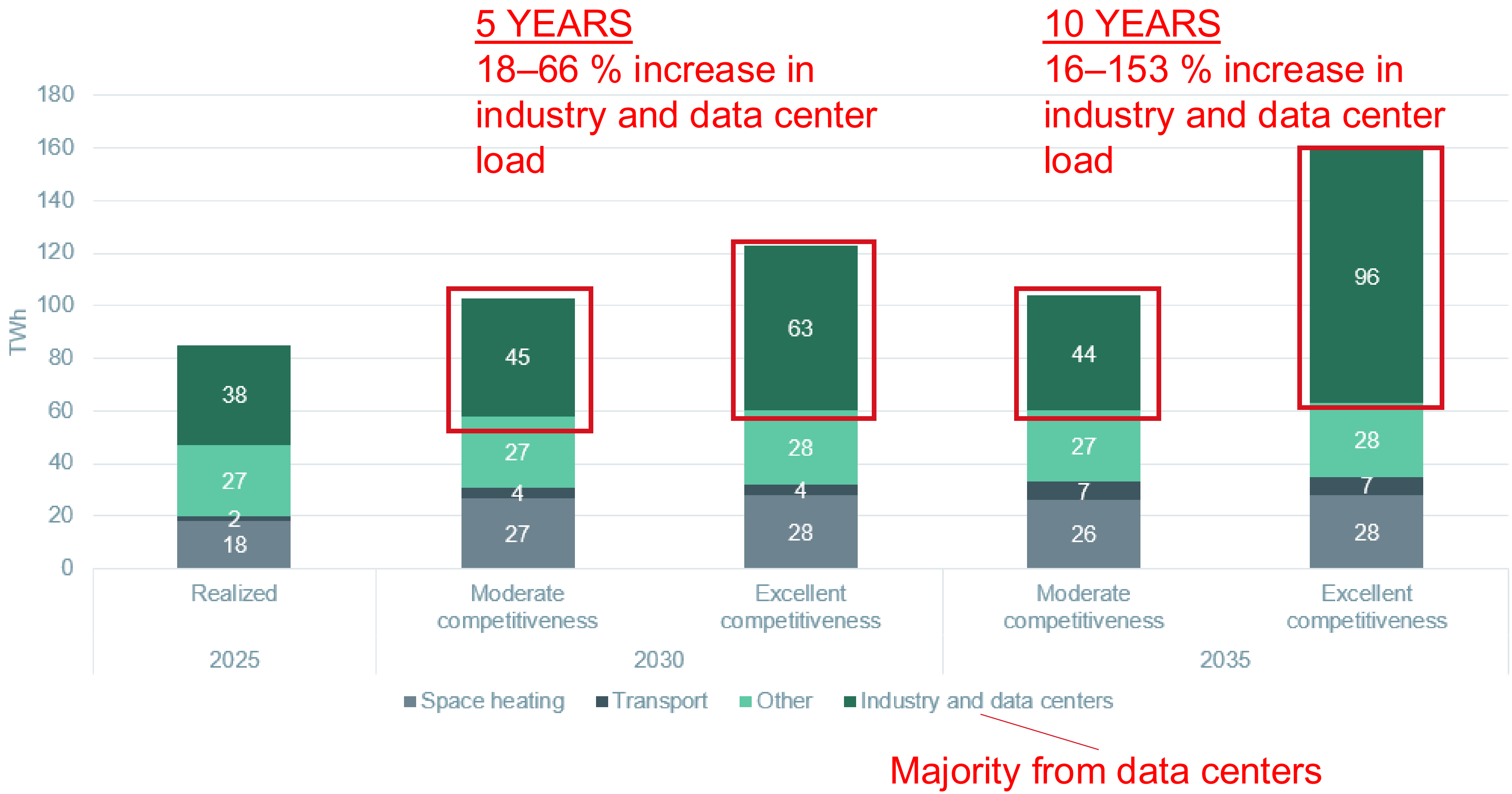
**The large number of enquiries is an indication of Finland's solid ability to compete in green transition investments.**

Peak consumption at the moment: **16 GW**

Consumption enquiries 2024–2025: **54 GW**

# Finland has promising opportunities to compete for green transition investments – Fingrid is preparing for a significant increase in electricity production and consumption in its grid planning

## Electricity consumption 2025

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# How to get models?

KJV2018

**FINGRID** 12 December 2018 1 (28)

**Grid Code Specifications for Demand Connections KJV2018**

**Contents**



9.11 Simulation model requirements

Simulations model of the power grid or demand facility do not, normally, need to be delivered as a separate simulation model. The connecting party shall deliver the required data on grid connection and on electrical equipment to be connected to at least 110 kV voltage level, such as substations, power lines, transformers and compensation devices and the electrical network's operational information. This data is included in the required data to be delivered as defined in Chapter 6.

Fingrid can require the connecting party to supply a simulation model based on separate consideration, to the extent outlined in article 21 of Regulation 2016/1388, if the demand facility includes nonlinear or uneven loads (e.g. converters or electric arc furnaces).

*KJV2026 coming this year including*

- *LVRT capability*
- *OVRT capability*
- *Phase jump performance*
- *Active power oscillation limits*
- *Post-fault active power recovery limit*

PSS/E model instructions

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**Fingrid modelling instruction for PSS/E models –**

Data Centers

13.3.2026

PSCAD model instructions

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**Fingrid modelling instruction for PSCAD models –**

Data Centers

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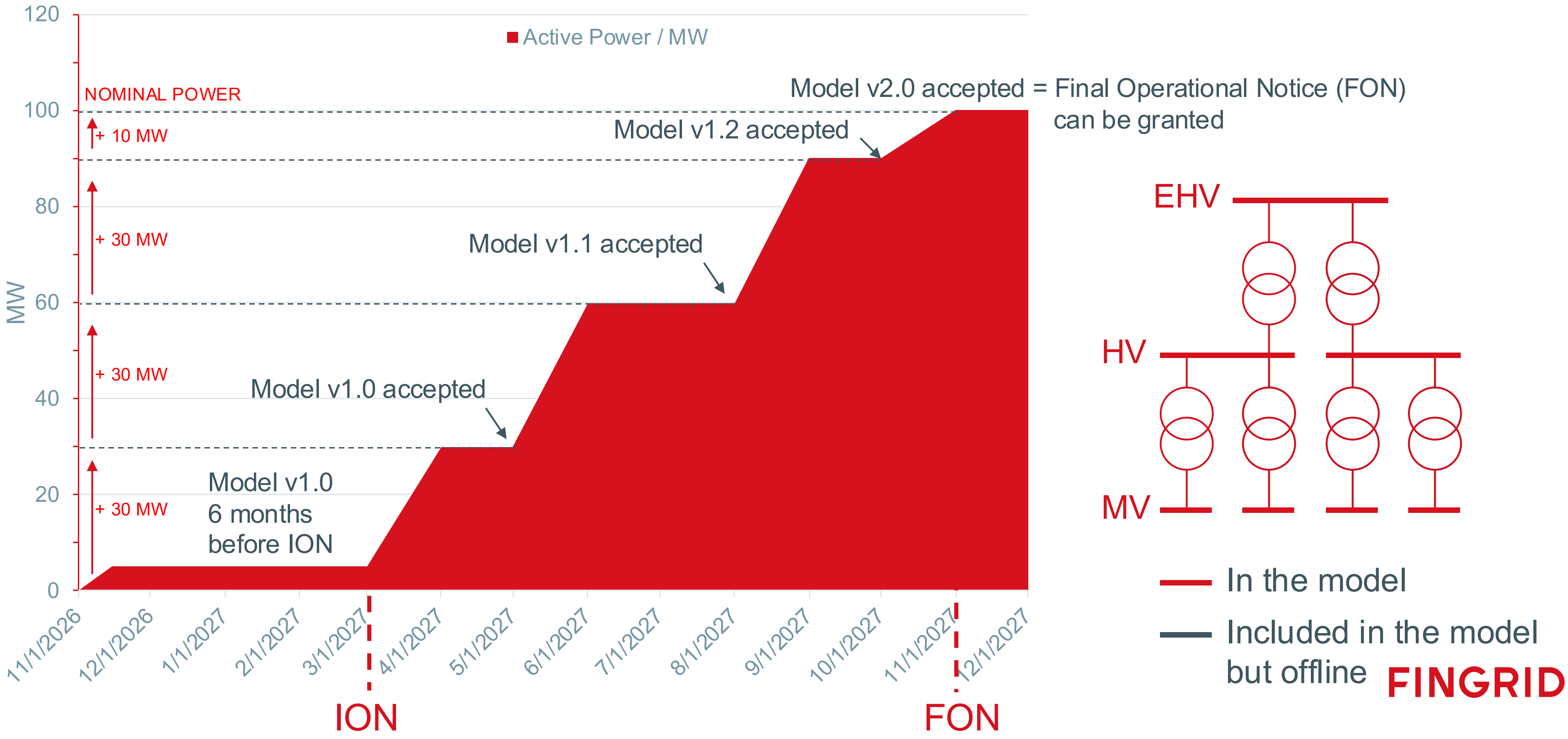
Demand facility of ≥30 MW like data center, electric boiler or P2G. What will be required?

1. Do not disconnect: FRT (0pu / 200ms) and post-fault active power recovery (90% / 1s)
2. Ability to limit active power consumption (at least 30%)
3. Do not oscillate
4. Controllability: readiness to take external control requests (such as power limitation)
5. Visibility: continuous measurements, simulation models
6. Specific study requirements: site specific additional requirements are possible

Flexibility!

| Requirement | | | Type of demand facility | | |
|---|---|---|---|---|---|
| | | | [illegible] | Type C 10 MW ≤Pmax,d< 30 MW | Type D Pmax,d ≥30 MW |
| Active power | 1 | Capability to limit active power consumption (continuous control / in steps) | Required | Required | Required |
| | 8 | Overvoltage ride-through OVRT (1.20 pu / 5 s) | Not required | [illegible] | Required |
| | 15 | Fingrid's direct control connection to demand facility | Not required | Not required | [illegible] |
| Specific study requirements | a | Subsyncronous interaction | Not required | Not required | If needed |
| | b | Geomagnetically induced currents | Not required | Not required | If needed |
| | c | Damping of power oscillations | Not required | Not required | If needed |
| | d | Low minimum short-circuit power at the connection point | Not required | Not required | If needed |
| | e | Converter interaction phenomena | Not required | Not required | If needed |
| | f | Dependency of the connected equipment from other equipment or connections | Not required | Not required | If needed |
| | g | Power quality | Not required | Not required | If needed |
| | h | Need for dynamic voltage control | Not required | Not required | If needed |

# Active Power ramp-up schedule vs. modelling

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# What to model?

- Key points of modelling
  - Principles of model aggregation. **Connectee shall propose how to aggregate** the ITE and auxiliary (mechanical) loads.
  - **Functional description** (operation of Demand Facility)
    - In normal continuous operation range (KJV2018: Figure 9.1 voltage- and frequency window)
    - Outside normal continuous operation range (KJV2018: Figure 9.1 voltage- and frequency window)
    - Load sharing and parallel production (behind POC, between same operator datacenters connected to Finnish grid)
- Intended use
  - Active and reactive power regulation
  - Fault situations: LVRT, HVRT, symmetric- and asymmetric faults, phase jumps
  - Power recovery: PFAPR
  - Oscillations
  - Subsynchronous interaction

# Fingrid's modelling instructions PSCAD

- Model shall accurately represent steady-state, dynamic, and transient behavior of the Data Center in
  - system side voltage and frequency disturbances **(data center internal faults not included!),**
  - interactions with system components and devices, including response to voltage and frequency oscillations etc. (full list in modelling instruction section 2.1)
- Time frame of interest is one (1) minute
  - **i.e. any logic or functionality that operates after 1 min of disturbance etc. can be left out**
- Converter models must be **average-value models (AVM)**
- UPS systems must be **vendor specific models** (with actual hardware code embedded to model)
- Simulation time step of 10 us must be supported
  - higher time step can and should be used if supported by the model. AVM models usually support 25 us
- Accurate representation in frequency range: 0.2 Hz to 2.5 kHz

# Fingrid’s modelling instructions PSS/E

- Model shall accurately represent **steady-state and dynamic behavior** of the Data Center during and after system faults, voltage disturbances and frequency disturbances seen at POC
- Time steps: 1..5 ms
- Time frame of interest: one (1) minute after disturbance
- Stable operation up to five (5) minutes
- Pay special attention to modelling of
  - under voltage ride-through (UVRT) characteristic and
  - post-fault active power recovery (PFAPR)
- **Library models** may be used if the chosen parameters represent **UVRT and PFAPR with sufficient accuracy**
  - If manufacturer-specific models are used, the details shall be discussed separately with Fingrid

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# How to make sure models work?

Any existing report accepted (if good enough)

**Appendix A: Guideline to model accuracy validation and documentation**

The model validation is comparison of type test results (and/or field measurements) of the actual equipment with the results from the corresponding simulation model under identical or similar test conditions.

If any individual active unit (UPS system, VFD, STATCOM, BESS, etc.) model has existing certificate or documentation of model validation, these shall be provided as part of the model documentation. If necessary, Fingrid can require the demand facility owner to perform the model validation of specific equipment/devices/systems/etc. If required, the scope of testing, model validation and documentation are provided separately by Fingrid.

If not available, new testing shall be agreed with Fingrid

Factory test results

Validation

PSCAD-model

Validation

PSS/E-model

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